Lab Exercise 17- Creating Secrets in Kubernetes

Creating secrets in Kubernetes is an essential practice for managing sensitive information such as passwords, OAuth tokens, and SSH keys. Let's go through a lab exercise that demonstrates how to create different types of secrets in Kubernetes.

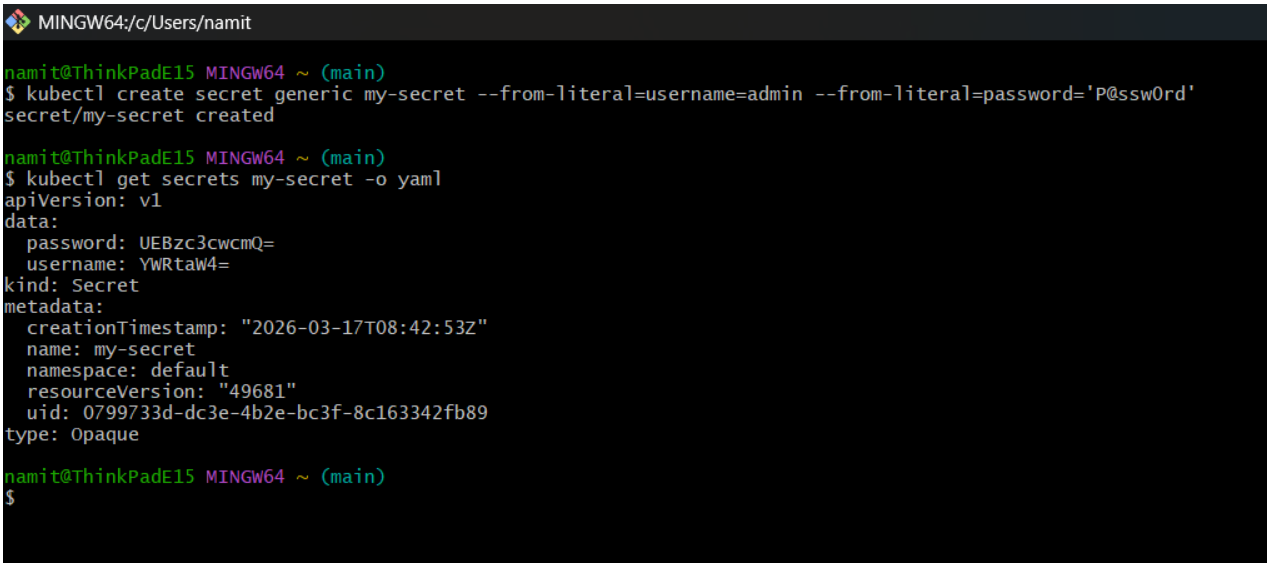
Step 1: Create a Generic Secret

Create a secret from literal values:

```
kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password='P@ssword'
```

Verify the secret:

```
kubectl get secrets my-secret -o yaml
```



The output will be base64 encoded. To decode, you can use:

```
echo -n “P@ssword” | base64
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ echo -n “P@ssw0rd” | base64
4oCcUEBzc3cwcmTigJ0=

namit@ThinkPadE15 MINGW64 ~ (main)
$ |
```

Step 2: Create a Secret from a File

Create a file with your sensitive data:

```
echo -n 'my-username' > ./username.txt



echo -n 'my-password' > ./password.txt
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ echo -n 'my-username' > ./username.txt

namit@ThinkPadE15 MINGW64 ~ (main)
$ echo -n 'my-password' > ./password.txt

namit@ThinkPadE15 MINGW64 ~ (main)
$
```

Create a secret from the file:

```
kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt
```

Verify the secret:

```
kubectl get secrets my-file-secret -o yaml
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt
secret/my-file-secret created

namit@ThinkPadE15 MINGW64 ~ (main)
$ kubectl get secrets my-file-secret -o yaml
apiVersion: v1
data:
  password: bXktcGFzc3dvcmQ=
  username: bXktdXNlcm5hbWU=
kind: Secret
metadata:
  creationTimestamp: "2026-03-17T08:45:24Z"
  name: my-file-secret
  namespace: default
  resourceVersion: "49880"
  uid: d1015666-df4f-4a23-995e-0f658f52afe2
type: Opaque

namit@ThinkPadE15 MINGW64 ~ (main)
$
```

Step 3: Create a Docker Registry Secret

Create a Docker registry secret:

```
kubectl create secret docker-registry my-registry-secret --docker-username=<your-username> --docker-password=<your-password> --docker-email=<your-email> --docker-server=<your-docker-server>
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ kubectl create secret docker-registry my-registry-secret --docker-username=namit53 --docker-password=dckr_pat_3UU8pirgpXMnjQ1y-SYDMHc9QMo --docker-email=namitra
ocker.io/v1/
secret/my-registry-secret created

namit@ThinkPadE15 MINGW64 ~ (main)
$
```

Verify the secret:

```
kubectl get secrets my-registry-secret -o yaml
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ kubectl get secrets my-registry-secret -o yaml
apiVersion: v1
data:
  .dockerconfigjson: eyJhdXRocyI6eyJodHRwczovL2luZGV4LmRvY2tlci5pby92MS8iOnsidXNlcm5hbWUiOiJuYW1pdDUzIiwicGFzc3dvcmQiOiJkY2tyX3BhdF8
29tIiwiYXV0aCI6ImJtRnRhWFExTXpwalkydHlYM0JoZEY4elZWVTRjR2x5WjNCWVRXNXFVVEY1TFZOWlJFMUlZemxSVFc4PSJ9fX0=
kind: Secret
metadata:
  creationTimestamp: "2026-03-17T08:54:51Z"
  name: my-registry-secret
  namespace: default
  resourceVersion: "50631"
  uid: d160fbb5-107f-4c85-9539-c14162c0f1d9
type: kubernetes.io/dockerconfigjson

namit@ThinkPadE15 MINGW64 ~ (main)
$
```

Step 4: Create a Secret Using a YAML Manifest

Create a YAML file for the secret:

Create a file named **my-secret.yaml** with the following content:

```
apiVersion: v1
kind: Secret
metadata:
  name: my-manifest-secret
type: Opaque
data:
  username: YWRtaW4=    # base64 encoded value of 'admin'
  password: UEBzc3cwcmQ=  # base64 encoded value of 'P@ssword'
```

Apply the YAML file:

```
kubectl apply -f my-manifest-secret.yaml
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ kubectl apply -f my-secret.yaml
secret/my-manifest-secret created

namit@ThinkPadE15 MINGW64 ~ (main)
$
```

Verify the secret:

```
kubectl get secrets my-manifest-secret -o yaml
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ kubectl get secrets my-manifest-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","data":{"password":"UEBzc3cwcmQ=","username":"YWRtaW4="},"kind":"Secret","metadata":{"annotations":{},"name":"my-manifest-secret","namespace":"default"},"type":"Opaque"}
  creationTimestamp: "2026-03-17T08:57:32Z"
  name: my-manifest-secret
  namespace: default
  resourceVersion: "50845"
  uid: 29bf4232-e3ba-4819-a252-babe23ac5f32
type: Opaque

namit@ThinkPadE15 MINGW64 ~ (main)
$
```

Step 5: Use Secrets in Pods

Create a pod that uses the secret:

Create a file named **pod-with-secret.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
    image: nginx
    env:
    - name: USERNAME
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: username
    - name: PASSWORD
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: password
```

Apply the YAML file:

```
kubectl apply -f pod-with-secret.yaml
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ notepad pod-with-secret.yaml

namit@ThinkPadE15 MINGW64 ~ (main)
$ kubectl apply -f pod-with-secret.yaml
pod/pod-with-secret created

namit@ThinkPadE15 MINGW64 ~ (main)
$
```

Verify the pod is running and the secret is being used:

```
kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ kubectl exec pod-with-secret -c mycontainer -- printenv | grep -E 'USERNAME|PASSWORD'
USERNAME=admin
PASSWORD=P@ssw0rd

namit@ThinkPadE15 MINGW64 ~ (main)
$ |
```

Step 7: Clean Up

Delete the secrets:

```
kubectl delete secret my-secret
kubectl delete secret my-file-secret
kubectl delete secret my-registry-secret
kubectl delete secret my-manifest-secret
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ kubectl delete secret my-secret
kubectl delete secret my-file-secret
kubectl delete secret my-registry-secret
kubectl delete secret my-manifest-secret
secret "my-secret" deleted
secret "my-file-secret" deleted
secret "my-registry-secret" deleted
secret "my-manifest-secret" deleted

namit@ThinkPadE15 MINGW64 ~ (main)
$
```

Delete the pod:

```
kubectl delete pod pod-with-secret
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ kubectl delete pod pod-with-secret
pod "pod-with-secret" deleted

namit@ThinkPadE15 MINGW64 ~ (main)
$
```